Sedimentation and Vegetation-driven Morphodynamics in a Regulated Reservoir: A Modelling Study of Lake Seminole (Florida, US)

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Background & Motivation

Reservoir sedimentation is the continuous trapping of material transported by inflowing rivers. This accumulation represents a major challenge for storage capacity, navigation, and preservation of infrastructure (e.g., turbine components). In some cases, it leads to the formation of subaerial islands and deltaic environments analogous to coastal systems.

Lake Seminole (Florida, USA) is an artificial reservoir created in 1954 by the construction of the **Jim Woodruff Dam** at the confluence of the Chattahoochee, Flint, and Spring Creek rivers. It offers an ideal setting to examine how river regulation affects reservoir sedimentation. The lake has two main arms:

- **Chattahoochee arm**: fed by a heavily regulated river system, with **strong daily discharge fluctuation** due to energy production.
- **Flint arm**: fed by a minimally regulated reach.

Although reservoir operations keep water level fluctuation minimal, rapid sedimentation has led to the formation of a subaerial delta only in the Chattahoochee arm.

This study presents ongoing modelling of sedimentation processes in Lake Seminole. The goal is to understand how hydraulic conditions drive morphological change and control reservoir sedimentation. The results highlight the impact of upstream regulation on sedimentation patterns and delta evolution.

Methods & Data

We developed a **2D depth-averaged Delft3D-FM**[†] morphodynamic model of Lake Seminole.

Hydrodynamic boundary conditions:
- Chattahoochee: Discharge from USGS 02343801 gauge
- Flint: Discharge from USGS 02356000 gauge
- Spring Creek: Discharge from USGS 02357150 gauge
- Woodruff Dam: Water level from USGS 02357500 gauge

Sediment transport boundary conditions :
- Chattahoochee: SSC from rating curve*
- Flint: SSC from rating curve*
- Spring Creek: No timeseries

*SSC data retrieved from **Sentinel-2A/B MSI**

Vegetation and bottom sediments:
- Invasive vegetation distribution (**F**loating & **E**mergent **A**quatic **V**egetation and **Hydrilla**) from Sentinel-2A/B MSI.
- Bottom composition from grab samples. Three sediment fractions are defined: **sand**, **silt**, and **clay**.

Calibration/Validation data :
- Depth-averaged flow speed from Signature 1000 ADCP at fixed platforms S0, S1, S2 (Figure 1 Right).
- SSC from Teledyne ISCO water sampler at fixed platforms S0 and S2.

Calibration/Validation procedure :
- Flow speed measurements used to calibrate Manning coefficient and vegetation parameters using **OpenDA**[‡] during low and high flow.
- SSC measurements used to calibrate critical shear stress for erosion and erosion parameter. Traditional approach by testing several values.

Key analysis:
- Lagrangian particle tracking **dorado**[§] used to understand water routing and connectivity.
- Morphodynamic simulations during 2 scenarios: Flood month vs Regular month.

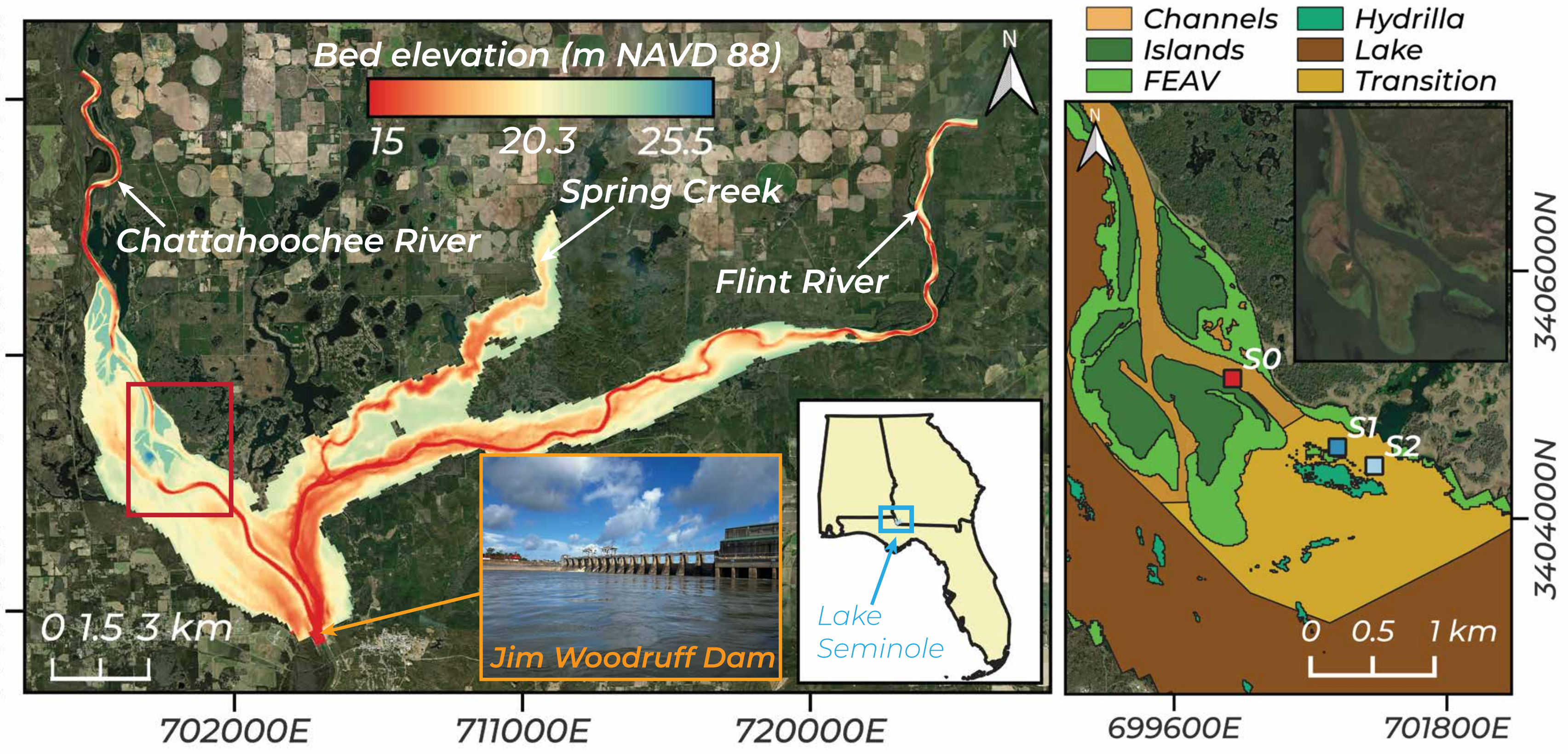


Figure 1. ***Left:*** Bed elevation of Lake Seminole. The bottom-right inset shows lake's location in the south-eastern US, while the bottom-center inset shows a view of the Jim Woodruff Dam (looking upstream). ***Right:*** Friction classes used in the Delft3D-FM model. The upper-right inset shows the Sentinel-2 scene used to derive the distribution of the invasive vegetation.

Key takeaway: Invasive vegetation affects the hydrodynamics of Lake Seminole. Model results (not pictured) demonstrate that measured flow velocities cannot be reproduced without including vegetation.

References:

[†] **Deltares**. (2025). D-Flow Flexible Mesh/D-Real Time Control/ D-Morphology User Manuals.

[‡] **OpenDA**. https://docs.openda.org/en/latest/index.html

[§] **Hariharan et al.**, (2020). dorado: A Python package for simulating passive particle transport in shallow-water flows. Journal of Open Source Software, 5(54), 2585, https://doi.org/10.21105/joss.02585

Nourishment & Routing

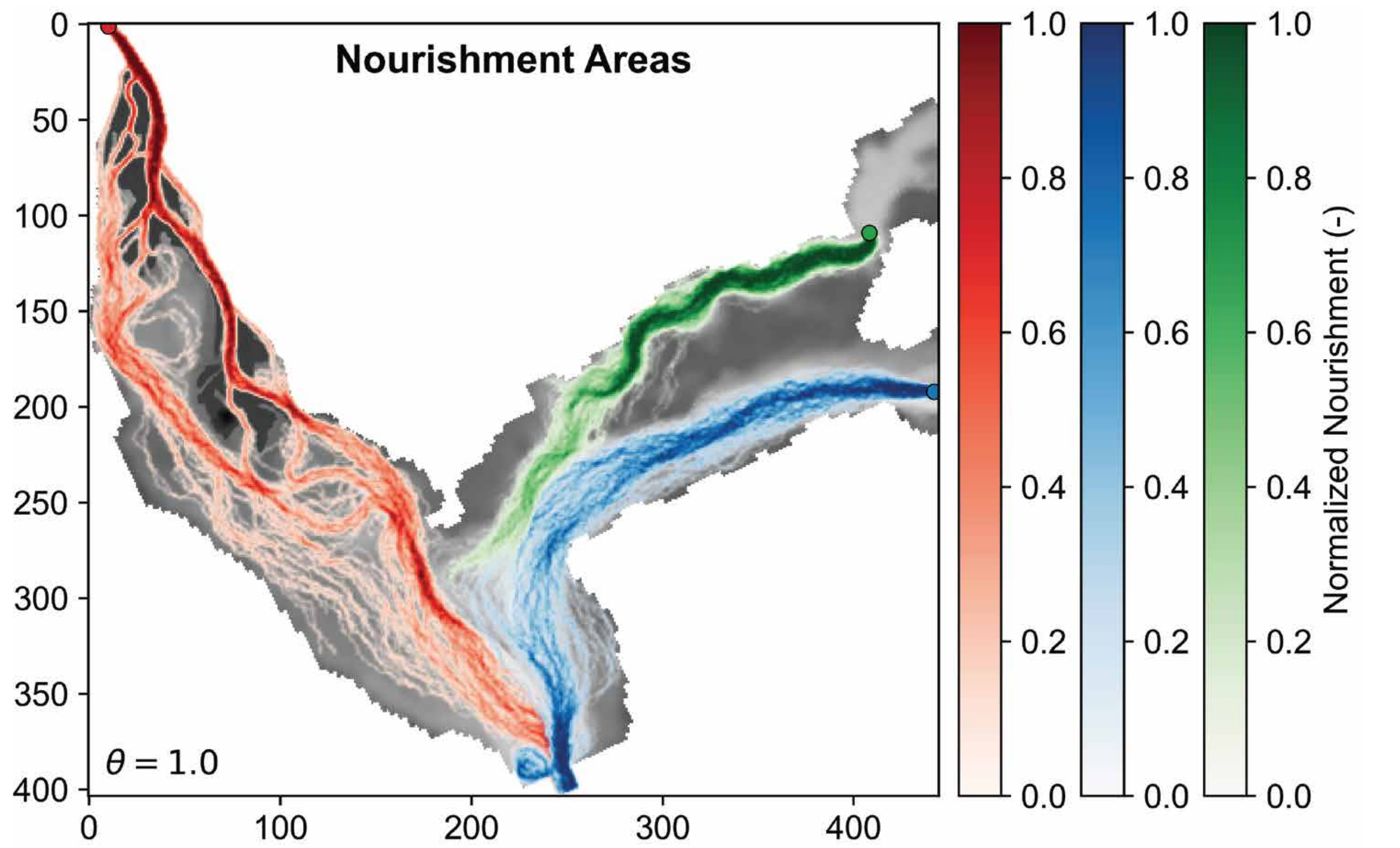


Figure 2. Nourishment areas from dorado. Particles are injected at three points (one per river reach). The normalized nourishment area shows the fraction of particles passing each location (1=all, 0=none). Paths are colored by river reach (e.g. blue=Flint River). Θ controls particle routing: Θ=1 sets a linear dependence on flow depth (i.e. Θ=1 is representative of water).

Key Takeaway: Particles paths show no mixing between the Flint and Chattahoochee waters even in the confluence area in the lower part of the reservoir.

Reservoir Sedimentation & Erosion

NET VOLUME CHANGE

$$\Delta V = \sum_{i=1}^{N} \Delta z_i A_i$$

N: number of cells
Δz: accumulation/erosion (m)
A: cell area (m²)

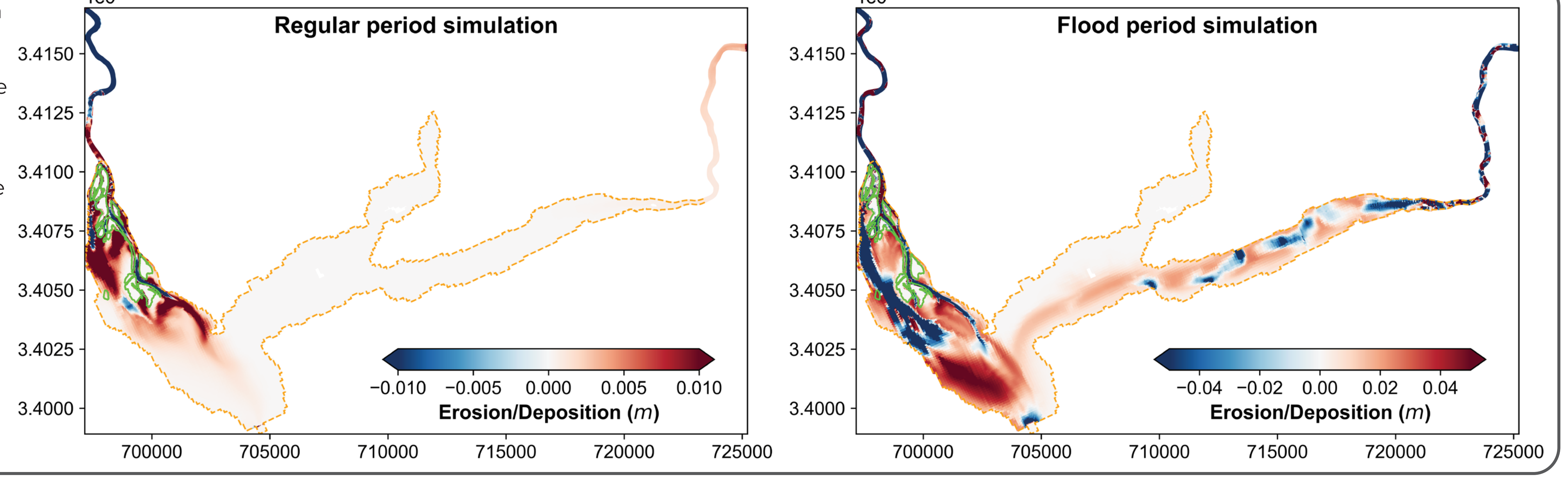
REGULAR PERIOD:
ΔV~80000 m³

FLOOD PERIOD:
ΔV~-200000 m³

Figure 4. Lake bed evolution during a Regular period (left) and a Flood period (right). The orange dashed line marks the lake area used to compute morphological change.

Key Takeaway: The simulations suggest that regular periods are net depositional, while periods characterized by large floods are net erosional.
IMPORTANT: Additional analysis (sediment balance) is needed to confirm this, as floods may redistribute rather than export the eroded sediment.

Wait - There's more!! I couldn't include everything here, so I created a GitHub repository for the project where you can check out the rest.



Effect of Water Management on Sediment Transport

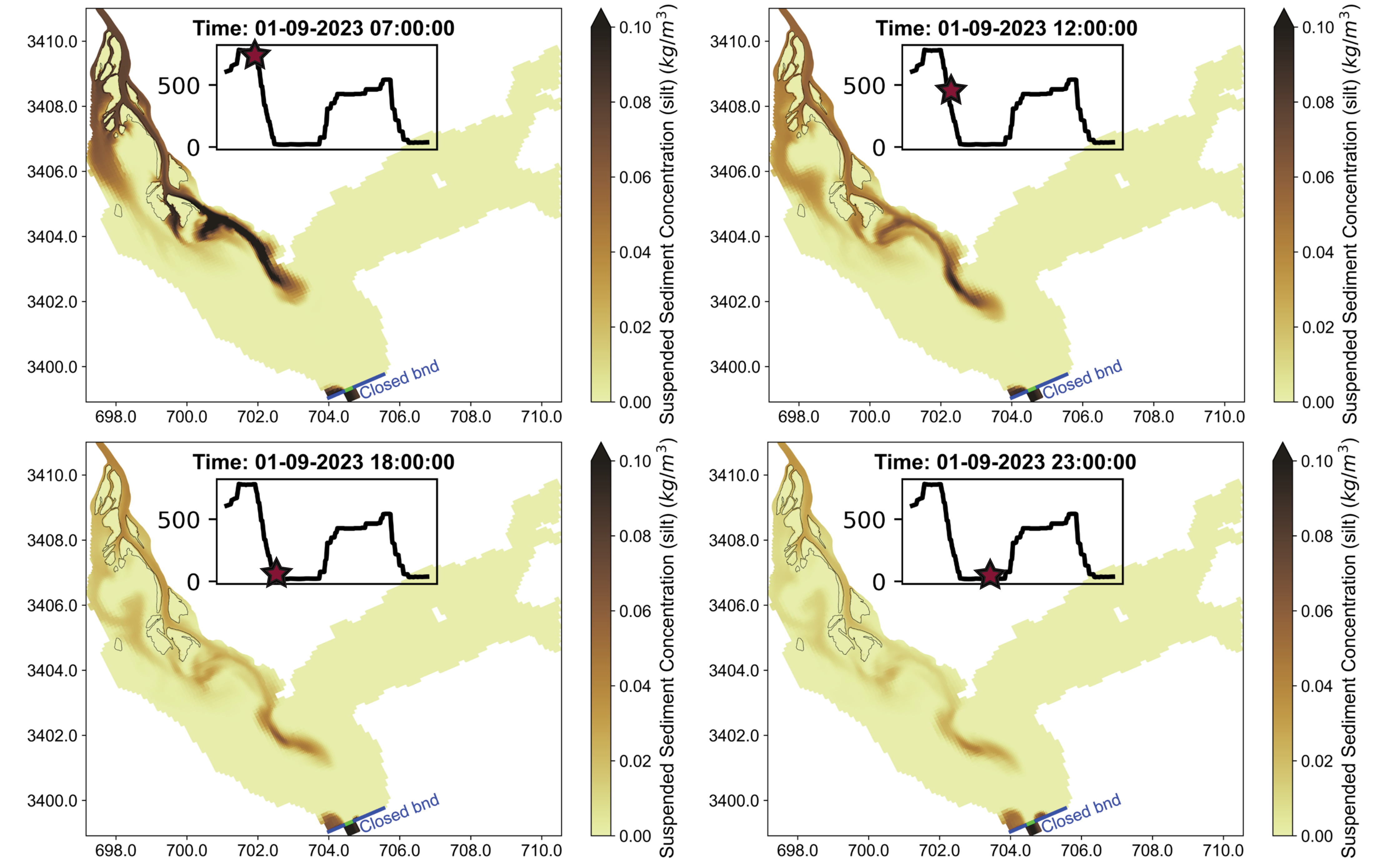


Figure 3. Four snapshots of silt SSC on 1 September 2023. The green segment marks the dam (the only model's outlet). SSC in the Flint is very low in these snapshots. Each panel includes an inset showing the discharge value (in m³/s). Only silt is shown; sand is resuspended only during floods and clay behaves as wash load.

Key Takeaway: Daily discharge fluctuations driven by hydropower redistribute cohesive sediment (silt and clay) throughout the reservoir during peaks. As discharge drops, sediment settles but is quickly remobilized by the next peak, producing a clear contrast between the fine sandy channels and the cohesive lake bottom.

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